# Linux.Mirai Incident Response - Student Briefing

## 1. Malware Overview

- Family / type: Linux.Mirai (Trojan)
- Aliases: Linux.DDoS.87, Linux.DDoS.89
- Behaviour class: DDoS botnet malware
- What it does: Linux.Mirai targets Linux systems, particularly IoT devices. It performs DDoS attacks by infecting devices and using them as part of a botnet. The malware collects sensitive data such as /etc/passwd, home directory listings, and other sensitive files, then exfiltrates them to a C2 server.
- Associated MITRE ATT&CK techniques:
- Persistence: T1547.001 (Registry Run Keys / Startup Folder)
- Command and Control: T1071 (Application Layer Protocol)
- Exfiltration: T1041 (Exfiltration Over C2 Channel)
- Associated CVEs: Not specified in this scenario.

## 2. Exercise Objective

- The learning goal of this cyber-range exercise is to investigate a compromised host infected with the Linux.Mirai trojan, identify its activities, and take appropriate measures to contain and remediate the threat.
- Difficulty level: Intermediate
- Cybersecurity approach: Defence-oriented
- NICE work role: Incident Response
- Successful trainee: By the end, a successful trainee will have demonstrated the ability to investigate a compromised host, identify suspicious services, determine network ports used for unauthorized communication, and identify persistence mechanisms.

## 3. Sandbox Topology & Malware Emulation

- Hosts:
- dist-server: 192.168.60.5
- victim: 192.168.60.10
- c2-server: 192.168.60.20
- router: 192.168.60.1
- Network layout:

```
[router] <---> [dist-server] <---> [victim] <---> [c2-server]
```

- Malware emulation:
- Service: dvrHelper.service (emulates the suspicious activity of collecting and exfiltrating data)
- Port: 5556 (used for communication with the C2 server)
- Persistence mechanism: dvrHelper.timer (ensures the service runs across reboots and process termination)
- Files:
- /usr/local/lib/.netd/harvester.sh (script for collecting and exfiltrating data)
- /var/lib/.netd/harvest.log (log file for the harvester script)
- /etc/systemd/system/dvrHelper.service (service configuration)
- /etc/systemd/system/dvrHelper.timer (timer configuration)
- Hosts:
- victim: Host running the emulated malware behaviour.
- c2-server: Host receiving the exfiltrated data.

## 4. Student Tasks

- ACCESS step:

- Credentials: Username: cyrano, Password: cyrano
- Start passkey: start
- TRAINING_LEVEL 1:
- Task: Investigate the running services and identify the service responsible for the suspicious activity of periodically collecting /etc/passwd, home directory listings, and sensitive files, then exfiltrating them to a C2 server.
- TYPE of artifact/flag expected: Service name.
- TRAINING_LEVEL 2:
- Task: Investigate the service and determine the network port associated with the unauthorized communication for exfiltrating data to a C2 server.
- TYPE of artifact/flag expected: Port number.
- TRAINING_LEVEL 3:
- Task: Determine which system component is responsible for keeping the identified service running across reboots and process termination.
- TYPE of artifact/flag expected: Timer name.

## 5. Solution Walkthrough & Flags

- TRAINING_LEVEL 1:
- Methodology:
    1. List all services on the system using systemctl list-units --type=service --all.
    2. Look for unfamiliar service names or misspellings of legitimate services.
    3. Check the service descriptions and startup behavior for any unusual activity.
    4. Identify the service responsible for the suspicious activity.
- Expected flag: dvrHelper.service
- TRAINING_LEVEL 2:
- Methodology:
    1. Use systemctl cat dvrHelper.service to inspect the service configuration.
    2. Pay particular attention to the ExecStart field, which specifies the path to the script launched by the service.
    3. Inspect the script to find the port number used for communication.
- Expected flag: 5556
- TRAINING_LEVEL 3:
- Methodology:
    1. List all timers on the system using systemctl list-timers --all.
    2. Look for timers associated with the identified service.
    3. Inspect the timer configuration using systemctl cat <timer>.
    4. Identify the timer responsible for the recurring execution of the service.
- Expected flag: dvrHelper.timer

## 6. Knowledge Gained

- Skills:
- Usage of systemctl for service and timer management.
- Investigating suspicious services and network ports.
- Identifying persistence mechanisms.
- Tools:
- systemctl
- ncat
- Concepts:
- Malware persistence mechanisms.
- Command and control (C2) communication.
- Data exfiltration techniques.
- NICE role: Incident Response

- MITRE techniques:
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